CS 4795 Proposal

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GitHub: https://github.com/q62ur/CS-4795-AI-Project

Title: *Learning-Enhanced Adversarial Agent for the Fox and Hounds Game*

Category: C – Game playing combined with intelligent/learning behavior

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This project aims to design and implement an intelligent game-playing agent for the board game *Fox and Hounds*. The game will be modeled as a state-space search problem in which each board configuration represents a state and each legal move represents an action.

An adversarial search algorithm (Minimax with Alpha-Beta pruning) will be used to allow the agent to select strategic moves against an opponent. A heuristic evaluation function will be developed to estimate the quality of a board position for both the fox and the hounds.

To extend beyond a fixed strategy, the agent will also include a learning component. The program will play repeated games against itself and adjust the weights of its heuristic evaluation function based on game outcomes. Through this self-play process, the agent is expected to improve its decision-making ability over time.

The performance of the learning agent will be compared with a basic non-learning agent. Metrics such as win rate, number of moves, and situations where the fox escapes or is trapped will be analyzed to evaluate the effectiveness of the learning approach.

Preliminary Bibliography

1. Academic paper on *Fox and Hounds* strategy and analysis. https://scholarworks.calstate.edu/downloads/1v53k187p
2. *Fox and Hounds* Rules and Discussion. Reddit. https://www.reddit.com/r/Ruleshorror/comments/1gn9iw0/fox_and_hounds/
3. *Fox and Hounds* Game Description and Mathematical Discussion. https://mathwithbaddrawings.com/wp-content/uploads/2020/11/Game-33-Fox-and-Hounds.pdf
4. *Fox and Hounds* Strategy and Analysis. Gilith Blog. https://gilith.wordpress.com/2018/11/26/fox-hounds/
5. *Fox and Hounds – Game Explanation.* YouTube video. https://www.youtube.com/watch?v=6VjNwmGAyzo
6. *Fox and Geese (Fox and Hounds)* Mobile Game Application. Google Play Store. https://play.google.com/store/apps/details?id=com.donkeycat.foxandgeese&hl=en_CA